

ORF309/MAT380

Random Walks I:

Definitions & First Step Analysis

(pp. 115-122)

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Agenda for today

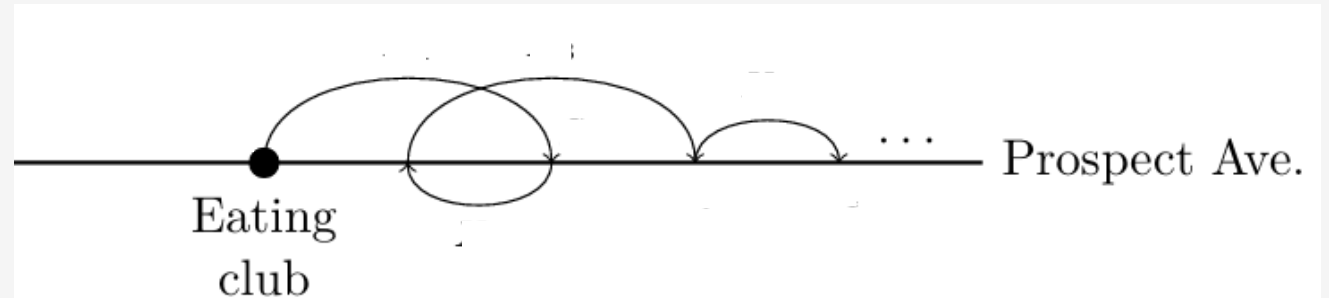
- Beginning of Random Walks with ...
 - Examples
 - Definitions
 - 3 important questions
- First Step Analysis

Random Walk Examples

- **Example 1:**

- You have just come out of eating club
- You make a random step to the **right**,
- Then a random step to the **left**,
- Then a random step to the **right**,
- Then a random step to the **right**,

How could we model this movement ??



- If your **objective** is to go back to your room and sleep, it is **uncertain** when (or if ever) you will achieve it

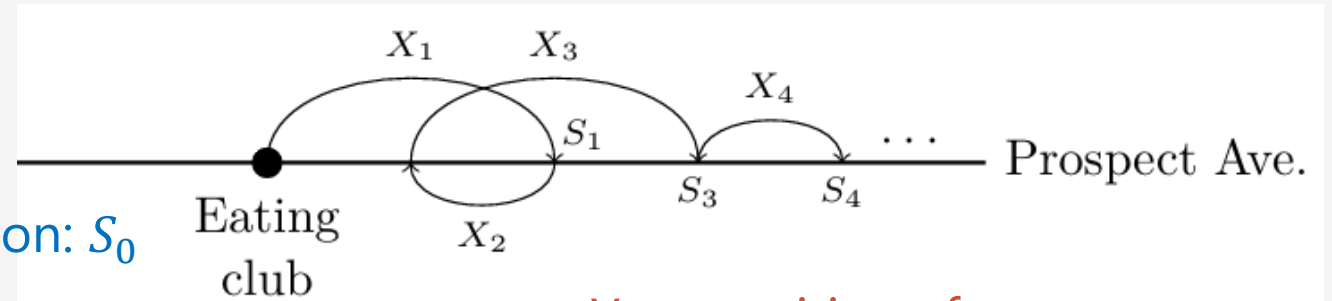
Random Walk Examples

- **Example 1:**

Let's try to model this randomness!

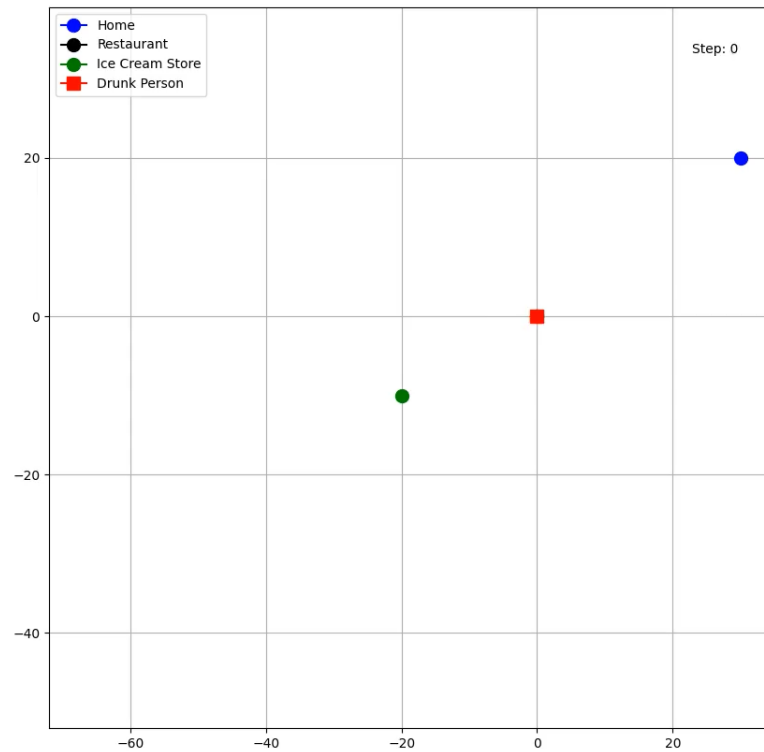
Your random steps: $X_1, X_2, X_3, \dots \rightarrow$ i.i.d.

Your initial position: S_0



Your position after n steps
 $S_n = S_0 + X_1 + X_2 + \dots + X_n$

Random Walk *in 2D*



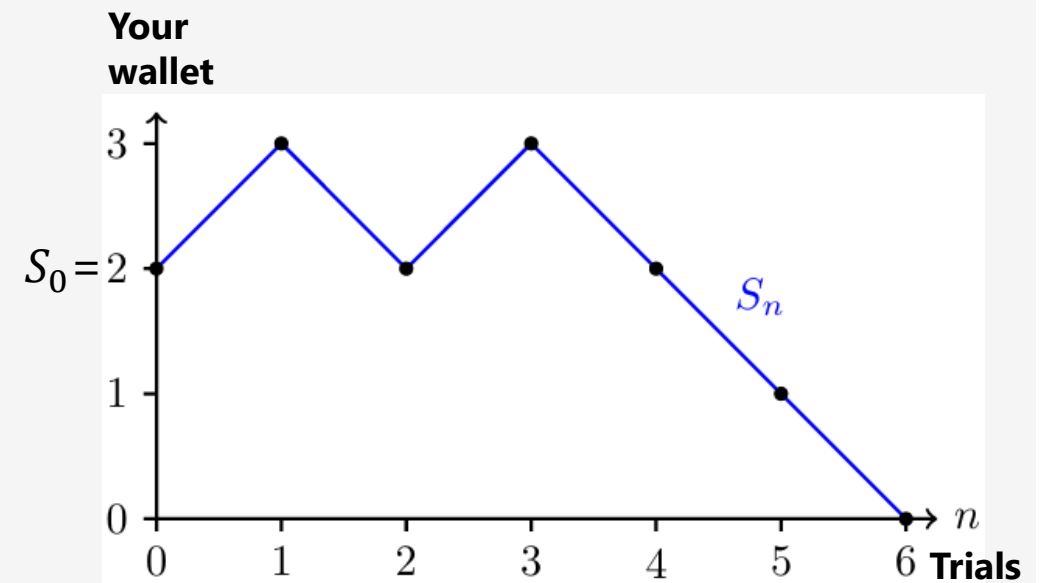
- **Example 2:**

- You and your best friend play the game:
 - Each player bets **\$1**
 - A **fair coin** is flipped (by a random player)
 - If **Heads** you win \$1 (you take your friend's \$1)
 - If **Tails** your friend wins \$1 (she takes your \$1)
 - In your pocket you have: **\$2**
 - You play this game until either of you decides to stop.

- **Example 2:**

- Let's analyze the game:

- Initial state: $S_0 = 2$
- Profit-or-Loss in trial k : $X_k \in \{+1, -1\}$
- $P\{X_k = +1\} = P\{X_k = -1\} = 1/2$ (**fair coin**)
- $X_1, X_2, X_3, \dots$ are i.i.d. & independent of S_0
- **Total wealth after n trials: ???**

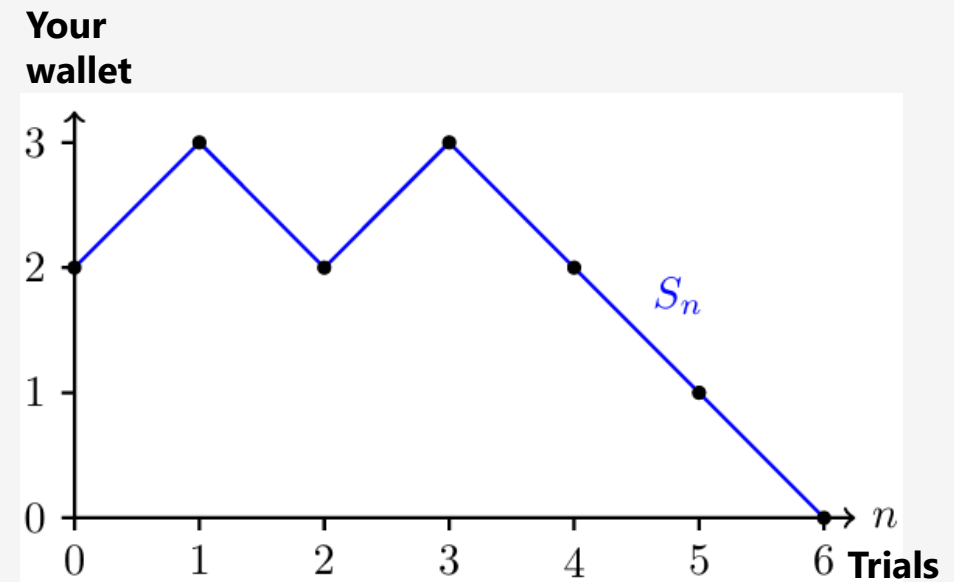


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- **Total wealth after n trials:** $S_n = S_0 + X_1 + X_2 + \dots + X_n$



-
- **Definition:** A **random walk** $\{S_n\}_{n \geq 0}$ is a random process whose k -th element is given by $S_k = S_0 + X_1 + X_2 + \cdots + X_k$, where $X_1, X_2, \dots$ are i.i.d. random variables and **independent** of the initial state S_0 .

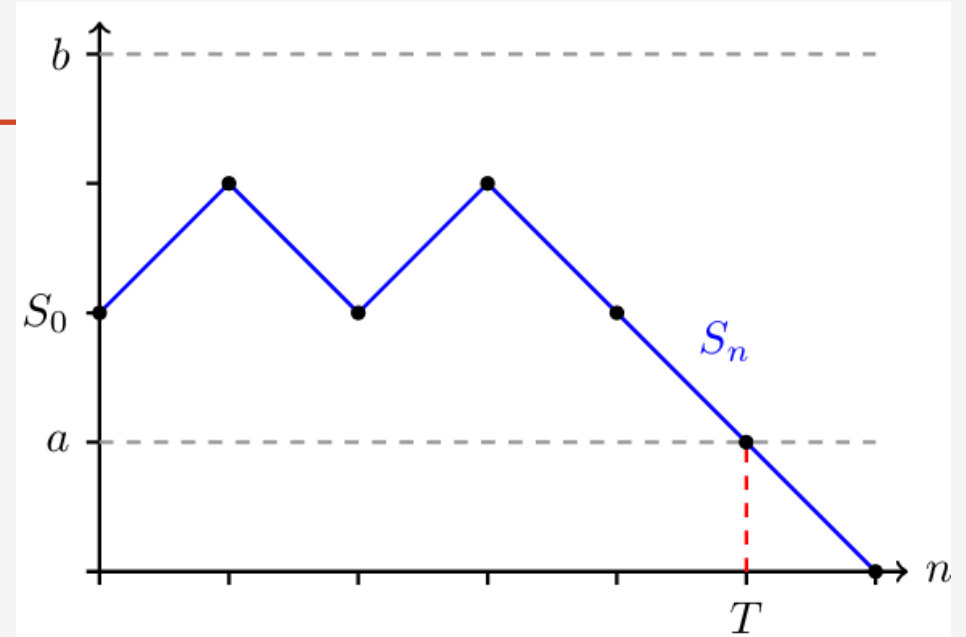
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 - **Definition:** A random walk $\{S_n\}_{n \geq 0}$ is called **simple** if S_0 is **integer** and the steps X_k can only take **± 1 values**.
 - **Definition:** A random walk $\{S_n\}_{n \geq 0}$ is called **symmetric** if it is simple and $P\{X_k = +1\} = P\{X_k = -1\} = 1/2$

Hitting times

Hitting times

- Consider a **simple symmetric** RW $\{S_n\}_{n \geq 0}$
- Define two integers a, b such that $a \leq S_0 \leq b$
 - a and b form the **barriers** of the RW
- The **hitting time** T of the RW is **the first time (trial) it touches** a or b
 - $T = \min\{n \geq 0 : S_n = a \text{ or } S_n = b\}$



- **Example 3:**

- You play a gambling game, and your initial position is $\$0 \leq S_0 \leq \100
- You stop when $S_n = \$0$ (you **bankrupt**) or $S_n = \$100$ (you reach your **goal**)
- **Question 1**: will the game end (hit any barrier), or go on for ever?
- **Question 2**: if it ends, what is the probability you will reach your goal?
- **Question 3**: on average, how long will it take for the game to end?

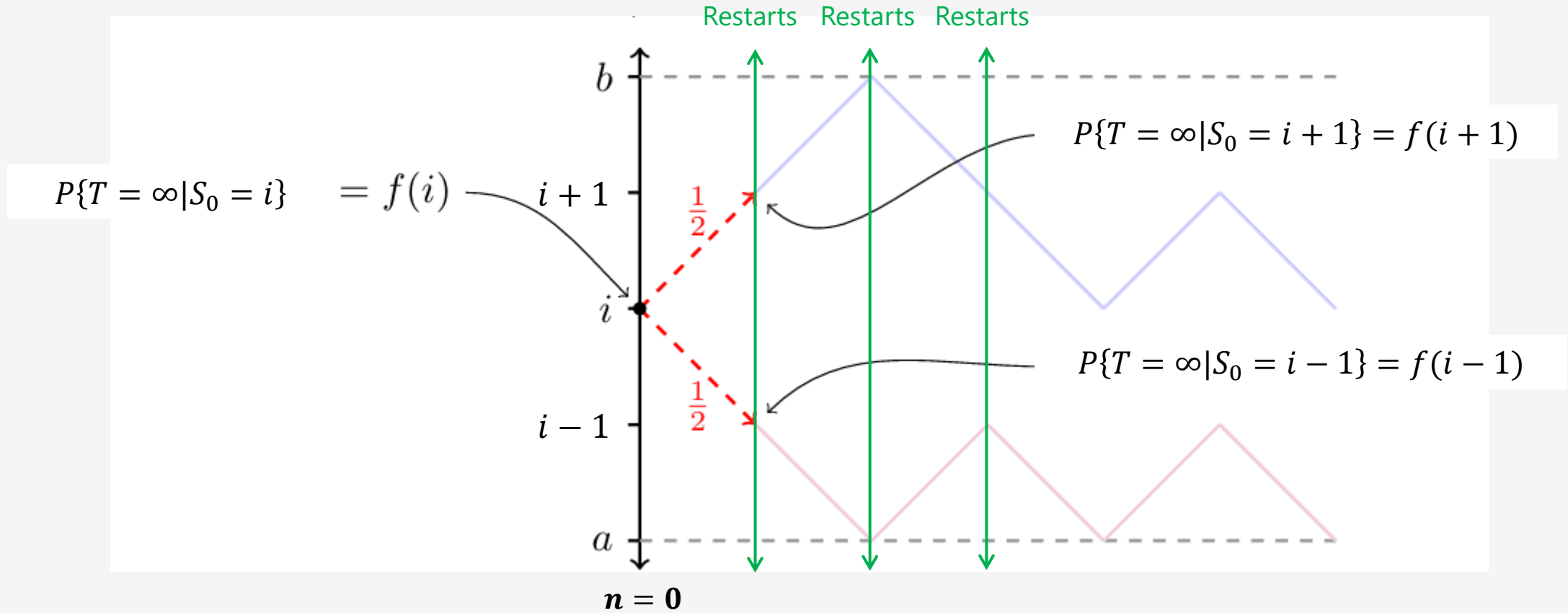
Q1: Will we ever hit a barrier?

- To answer this question, we will develop a very powerful technique called:
First Step Analysis.
- It is **easier** to find the probability to the question: **we will never hit a barrier, given that we started at position i**

$$f(i) = P\{T = \infty | S_0 = i\}, \quad a \leq i \leq b$$

- **Note**: this probability is a **function** of the **initial position i**

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- Let's analyze the case when the starting position is not on any of the barriers, that is $a < i < b$
 - The RW is **simple and symmetric** $\Rightarrow$
 - $X_i = +1$ or $X_i = -1$ with $P\{X_i = +1\} = P\{X_i = -1\} = 1/2$
 - $X_i \perp\!\!\!\perp X_j$
 - Because all steps X_i are **i.i.d.** the RW **restarts every time it takes a step**
 - $f(i+1)$ is the probability that the RW will never hit a barrier if it starts at position $i+1$



Slightly different perspective from the notes; focus on the restarts

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- **Let's put all this theoretical analysis in equations:**

$$\begin{aligned} f(i) &= P\{T = \infty | S_0 = i\} = \\ &= P\{T = \infty | S_0 = i, S_1 = i + 1\} P\{S_1 = i + 1 | S_0 = i\} + \\ &\quad + P\{T = \infty | S_0 = i, S_1 = i - 1\} P\{S_1 = i - 1 | S_0 = i\} \end{aligned}$$

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-
- $P\{T = \infty | S_1 = i + 1, S_0 = i\} =$
restart
 - $P\{T = \infty | S_1 = i - 1, S_0 = i\} =$

- $P\{T = \infty | S_1 = i + 1, S_0 = i\} = P\{T = \infty | S_{\textcolor{red}{0}} = i + 1\} = f(i + 1)$

restart

- $P\{T = \infty | S_1 = i - 1, S_0 = i\} = P\{T = \infty | S_{\textcolor{red}{0}} = i - 1\} = f(i - 1)$

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restart

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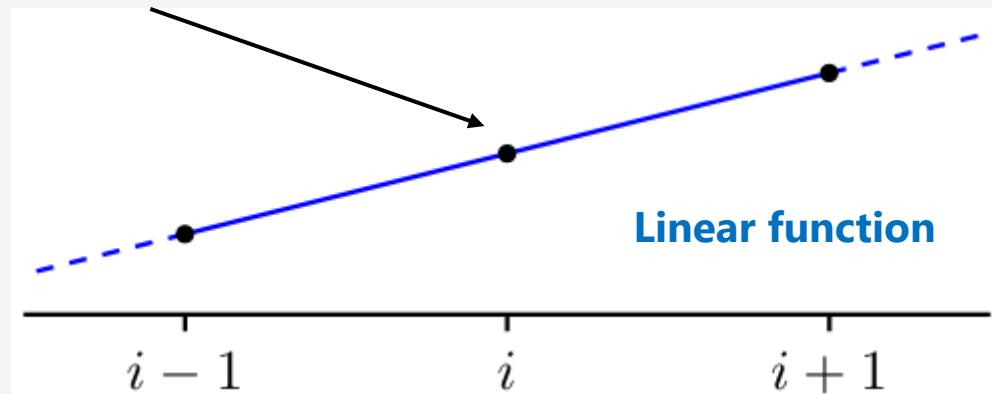
- Combining all these equations together we have

$$f(i) = \frac{1}{2}f(i + 1) + \frac{1}{2}f(i - 1), \quad a < i < b$$

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restart
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 - Combining all these equations together we have

$$f(i) = \frac{1}{2}f(i + 1) + \frac{1}{2}f(i - 1), \quad a < i < b$$

Possible plot of $f(i)$



But what happens at the boundaries??

-
- **What happens on the boundaries:** $f(a) = f(b) = ??$

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- **What happens on the boundaries:** $f(a) = f(b) = 0$

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 - So, we have a **linear function** that is **0 at the boundary** of the interval $[a, b]$
 - The function $f(i) = 0$ for all $a \leq i \leq b$
 - We have shown

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 - So, we have a **linear function** that is **0 at the boundary** of the interval $[a, b]$
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- We have shown

$$f(i) = P\{T = \infty | S_0 = i\} = 0, \quad a \leq i \leq b$$

- And hence $P\{T < \infty | S_0 = i\} = 1$, $a \leq i \leq b$
- **Hence, the RW will eventually hit a barrier regardless of its starting point**

Q2: Which boundary we will hit

- We now want to compute the **probability that** **we hit b** **before** **we hit a**

What event are we looking for? $\{ ? \mid ? \}$

Q2: Which boundary we will hit

- We now want to compute the **probability that we hit b before we hit a**

$$r(i) = P\{S_T = b | S_0 = i\} , \quad a \leq i \leq b$$

Q2: Which boundary we will hit

- We now want to compute the **probability that we hit b before we hit a**

$$r(i) = P\{S_T = b | S_0 = i\}, \quad a \leq i \leq b$$

- In the **Gambling game** this is equivalent to computing the probability for you to play till your **opponent is bankrupt**
- Working similarly as in the previous question (Q1) we have
 - $r(a) = 0$ and $r(b) = 1$ (**WHY?**)

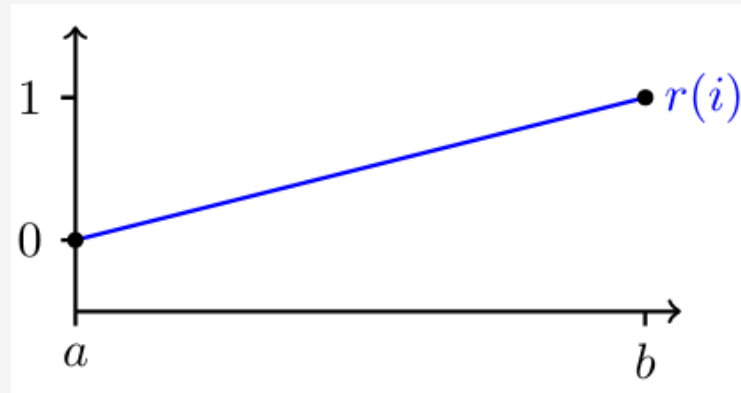
- $r(i) = \frac{1}{2}r(i+1) + \frac{1}{2}r(i-1), \text{ for all } a < i < b$

Linear function

How does its plot look like?

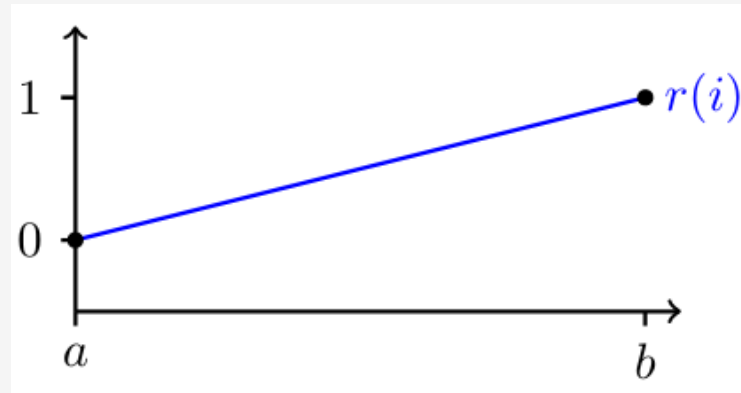
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- Analyzing the linear equations of $r(i)$ we can conclude that it is a straight line of the following form:



- The general formula is ??

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- The general formula is

$$r(i) = P\{S_T = b | S_0 = i\} = \frac{i - a}{b - a}, \quad a \leq i \leq b$$

- **Example:**

- Let's consider our original gambling example:
 - Your **aim** now is to play until you reach your **goal** (\$100), or you go **bankrupt** (\$0)
 - So, our **boundaries** are $a = 0$ and $b = 100$
 - We want to compute the probability:

$$r(i) = P\{S_T = 100 | S_0 = i\} = ??$$

- **Example:**

- Let's consider our original gambling example:
 - Your **aim** now is to play until you reach your **goal** (\$100), or you go **bankrupt** (\$0)
 - So, our **boundaries** are $a = 0$ and $b = 100$
 - We want to compute the probability:

$$r(i) = P\{S_T = 100 | S_0 = i\} = \frac{i - 0}{100 - 0} = \frac{i}{100} , \quad 0 \leq i \leq 100$$

- **Example:**

- Let's consider our original gambling example:
 - Your **aim** now is to play until you reach your **goal** (\$100), or you go **bankrupt** (\$0)
 - So, our **boundaries** are $a = 0$ and $b = 100$
 - We want to compute the probability:

$$r(i) = P\{S_T = 100 | S_0 = i\} = \frac{i - 0}{100 - 0} = \frac{i}{100}, \quad 0 \leq i \leq 100$$

- So, the more money you have (your initial position is closer to \$100) the probability of achieving more wealth is higher, while the poorer you are (your initial position is closer to \$0) more likely it is to bankrupt

- **Example (cont.):**

- This gambling game **does not seem to be fair...** BUT let's check if this is actually true?
- Let's compute the **expected money** we will have when the game reaches any of the boundaries
You play the game many times

$$E[S_T | S_0 = i] =$$

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- This gambling game **does not seem to be fair...** BUT let's check if this is actually true?
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You play the game many times

$$E[S_T | S_0 = i] = 0 \times P\{S_T = 0 | S_0 = i\} + 100P\{S_T = 100 | S_0 = i\} = :$$

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$$E[S_T | S_0 = i] = 0 \times P\{S_T = 0 | S_0 = i\} + 100P\{S_T = 100 | S_0 = i\} = 100 \frac{i}{100} = i$$

- Hence on average, the amount of money you will have when you stop playing is the same as that when you started
- The same is also true for your opponent
- **So, the game is (at least mathematically) fair!**
